Short Paper Track

Unlink to Unlearn: Simplifying Edge Unlearning in GNNs

Jiajun Tan, Fei Sun, Ruichen Qiu, Du Su, Huawei Shen

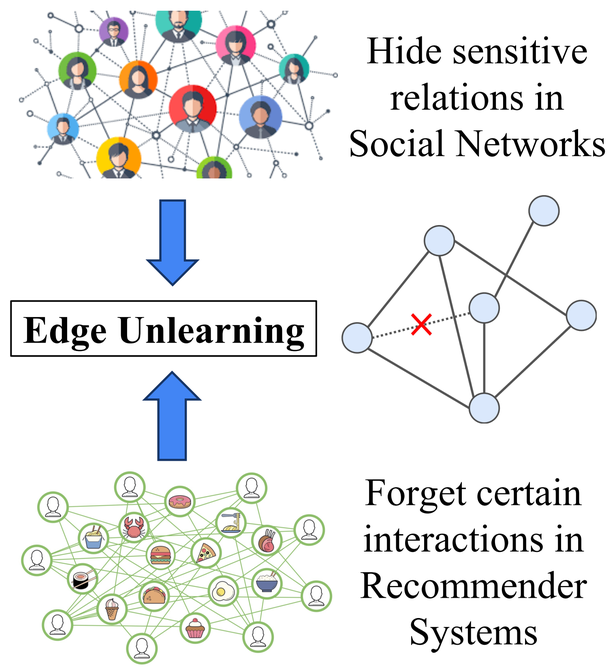
CAS Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences
University of Chinese Academy of Sciences

Paper

Code

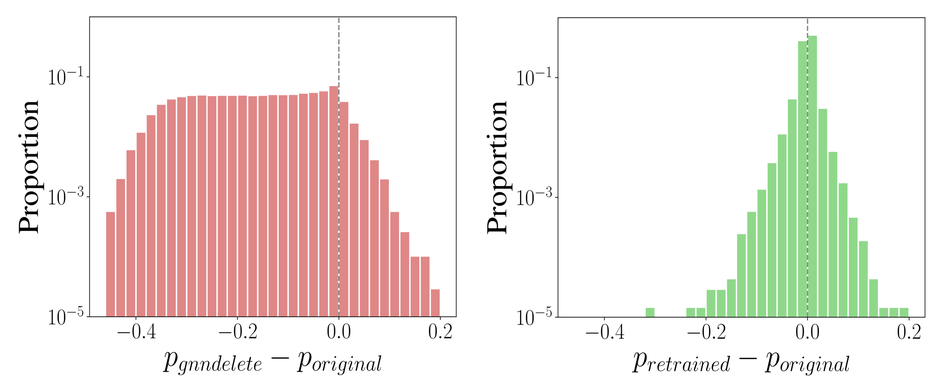
Introduction

- GNNs' training data may contain sensitive personal information, which is implicitly "remembered" within model parameters.
- Recent legislations, e.g., EU's GDPR and California's CCPA, have granted individuals with the *right to be forgotten*, enabling them to request elimination of their private data from online platforms.
- Machine unlearning has emerged to achieve quick and efficient removal of specific data from a trained model, rather than retraining from scratch.
- In this work, we focus on **edge unlearning**, which has a pivotal role in real-world applications for safeguarding edge privacy.



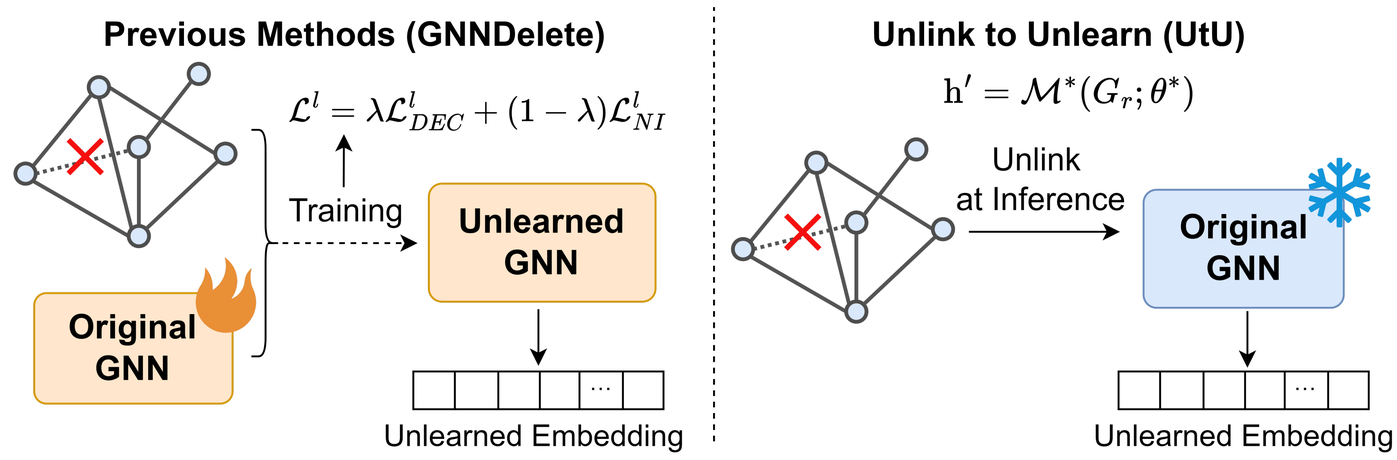
Approach

- **Over-forgetting:** Unlearning algorithms remove excessive information than necessary, causing significant decline in the retain set performance.
- Current leading method GNNDelete suffers from severe over-forgetting.
- We deprecate both DEC and NI loss in GNNDelete -- the source of over-forgetting, while achieving edge unlearning by unlinking forget edges.



Unlearn by GNNDelete Retrain from scratch

- GNNs' parameters are predominantly employed to transform node features after message passing, which does not involve edge utilization.
- Unlinking during inference is thereby enough to eliminate forget edges' influence from trained model by blocking message propagation paths.



Experiments and Results

Experimental results demonstrate that UtU acts on par with the retrained model with near-zero computational overhead.

- **Downstream Task Evaluation:** 👉 **Table 1**
- **Unlearning Efficacy Evaluation:** 👉 **Fig 1 & MIA** (👉paper)
- **Over-forgetting Evaluation:** 👉 **Fig 2**
  - Over-forgetting is measured by Δp: difference on prediction of retain edges between unlearned model and the retrained model.

Table 1: Link Prediction AUC

Dataset	Model	Retrain	GradAscent	GIF	GNNDelete	GNNDelete-NI	UtU
CoraFull	GCN	0.967	0.563	0.964	0.922	**0.967**	0.965
	GAT	0.963	0.766	0.926	0.934	0.947	**0.964**
	GIN	0.961	0.596	0.742	0.897	0.958	**0.960**
PubMed	GCN	0.970	0.375	0.924	0.934	0.968	**0.969**
	GAT	0.936	0.766	0.774	0.890	0.927	**0.933**
	GIN	0.939	0.545	0.842	0.887	0.938	**0.942**
CS	GCN	0.968	0.786	0.950	0.947	0.968	**0.970**
	GAT	0.963	0.846	0.941	0.943	0.958	**0.963**
	GIN	0.960	0.583	0.520	0.900	0.959	**0.960**
OGB-collab	GCN	0.985	0.406	0.971	0.925	0.981	**0.987**
	GAT	0.971	0.755	0.744	0.924	0.960	**0.971**
	GIN	0.925	0.683	0.500	0.805	0.912	**0.913**

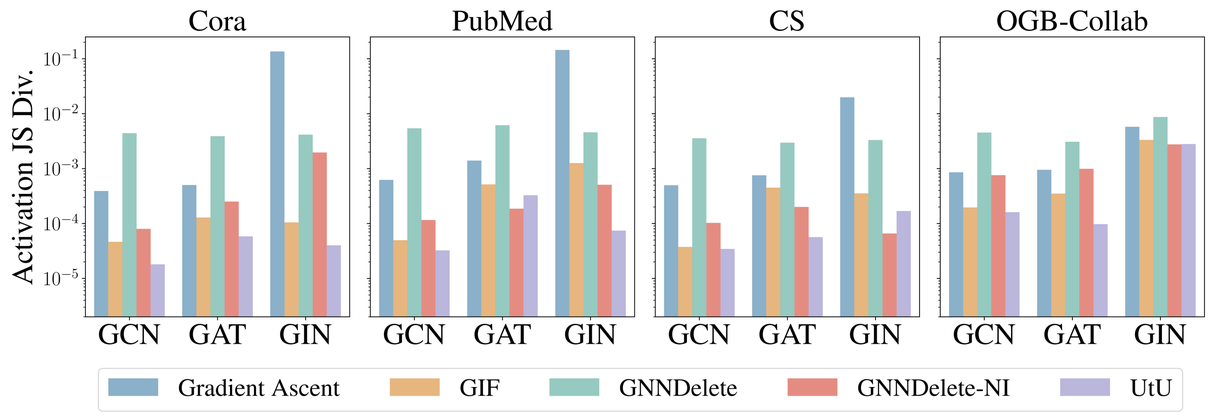


Fig 1: Output JS Divergence of forget edges

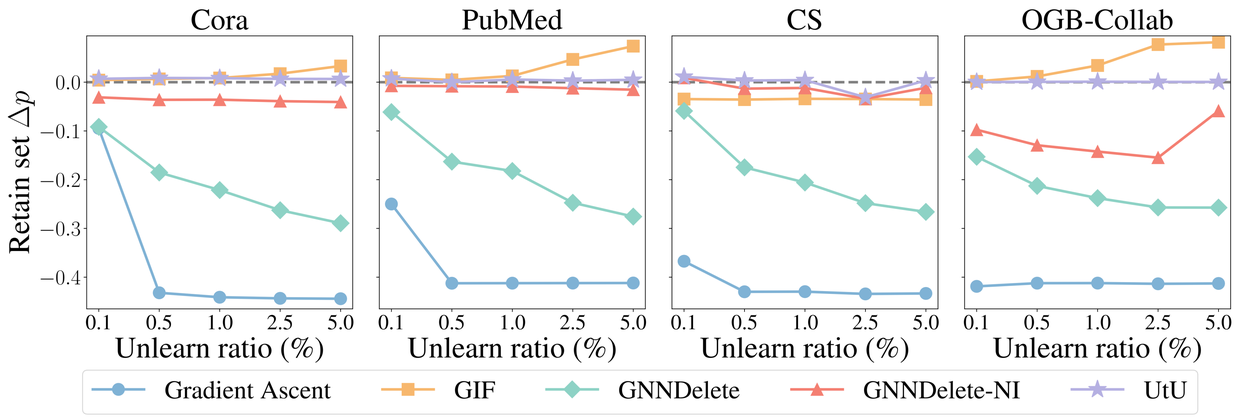


Fig 2: Over-forgetting comparison w.r.t forget size